CSCI 5707 (001, 883) Principles of Database Systems (Fall 2024)

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CSCI 5707: Principles of Database Systems
Fall 2024
Tuesday & Thursday, 2:30 pm to 3:45 pm, **Keller Hall 3-115**

Professor

Prof. Jaideep Srivastava (srivasta@cs.umn.edu)
Office Hours: Tuesday & Friday, 1:00 pm to 2:00pm, **ONLINE via zoom (link in Canvas)**

Teaching Assistant

Jiaxiang Tang (tang0836@umn.edu)
Office Hours: Monday, 2:30 pm to 4:40 pm, ONLINE via zoom:
https://umn.zoom.us/j/94593733142?pwd=VQVXvg8BOxW8UyHSXPveBDfWt8uhxs.1

Links to an external site.

Text Book

Database Management Systems by Raghu Ramakrishnan and Johannes Gehrke. Third Edition.

Book website (for slides used in class): http://pages.cs.wisc.edu/~dbbook/ (Links to an external site.)Links to an external site.

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Grading Criteria

Exam: 25% each exam (2 in total)

Homework: 5% each homework (4 in total); each team submits just one HW.

Project: 30%

- Team formation: 1%

- Project description: 3%
- Project check-in 1: 3%
- Project check-in 2: 3%
- Project report: 10%
- Project presentation: 10%

Important Notice: All homeworks and the project for this class are to be done in groups of 3-4. You are expected to work jointly, discuss with each other in the team, and make a single submission. The score on the submission will be the score of the whole team. So, please make sure you divide work, get started soon, and discuss with your teammates. We encourage you to form your teams immediately. If you haven't formed a team by Friday, September 8th, we will assign people into teams.

Class Schedule

Week	Chapter and Topic	Date	Additional Information
Week 1	1- Introduction	09/03	
	2- ER Model Conceptual Design	09/05	Form teams (4 per team) for HWs and the project
Week 2	3- Relational Model, Relational Algebra	09/10	HW1 OUT (Ch 1, 2, 3, 4, 5, 8)
	4- Relational Calculus, SQL Querying	09/12	
Week 3	5- SQL Querying	09/17	Submit project descriptions (1 page); should include title, team, objective, scope, approach

	5- SQL Querying 8 - Overview of Storage and Indexing	09/19	
Week 4	8 - Overview of Storage and Indexing	09/24	
	10 - Tree-structured Indexing	09/26	HW1 DUE (**1 day extension**)
Week 5	11 - Hash based indexing	10/01	HW2 OUT (Ch 10, 11, 12, 15, 16)
	12- Overview of Query Evaluation	10/03	
Week 6	15 – Query Optimization	10/08	
	16- Transaction Management · Overview · Concurrency control	10/10	
Week 7	16- Transaction Management · Recovery	10/15	HW2 DUE
	21- Security and Authorization	10/17	
Week 8	**EXAM 1**	10/22	**lectures through 10/17, HW1 and HW2**

	20- Physical DB Design, Tuning	10/24	Project check-in 1: Discuss project plan, list of references, etc. Groups will go in order, so Group1 first and Group 15 last. Each group will have 5 minutes. Be prepared with the following 3 slides: (i) The project and its scope, (ii) Progress to date, (iii) Plan for the rest of the semester. On optional 4th slide of 'issues to discuss' if there are any such issues. Zoom link available under the zoom tab.
Week 9	25- Data Warehousing and Decision support	10/29	
	26- Introduction to Data Science · Classification	10/31	HW3 OUT (Ch 20, 21, 25, 26)
Week 10	· Association Rule Analysis · Clustering	11/05	
	27- IR and XML Data · Text databases	11/07	
Week 11	27- IR and XML Data · Text databases	11/12	HW3 DUE

	Indexing and Query Text data	11/14	**Lecture on zoom (see link in zoom tab)**
Week 12	28- Spatial & Temporal Data	11/19	**Lecture by Jiaxiang Tang (TA)**
	Big Data Engineering · Hadoop, Map/Reduce	11/21	HW4 OUT (Ch 27, 28, Big Data)
Week 13	Hadoop, Map/Reduce	11/26	Project check-in 2: Discuss outline of report/presentation
		11/28	Thanksgiving (No Class)
Week 14	Amazon case study	12/03	
	Data Privacy	12/05	
	Project presentation **2:30pm - 4:15pm** **Groups 1 - 7, 12+3 minutes each**	12/10	HW4 DUE
Week 15	Project presentation **2:30pm - 4:15pm** **Groups 8 - 14, 12+3 minutes each**	12/12	
		12/13	Project report due

EXAM 2, 10:45am – 12:15pm Final exam times | Twin Cities One Stop Student Services (umn.edu)	12/19	**lectures from 10/20 to 12/10; HW3 and HW4**